

APPEALFORGE PAC

Denial-to-appeal evidence assembly for post-acute Medicare Advantage teams
2026-06-25

Build the product that makes winnable denials operationally
appealable.

Problem	Medicare Advantage post-acute denials are under-appealed because provider teams cannot assemble source-cited clinical evidence fast enough.
Wedge	SNF admission and continued-stay denials for Medicare Advantage plans.
Solution	Upload denial letter + clinical packet; parse denial; map evidence; identify gaps; generate human-reviewed appeal packet.
Buyer	Regional SNF/post-acute operators; VP Revenue Cycle, COO, VP Clinical Ops, Director of Admissions.
Codex objective	Build a polished local demo with seeded synthetic cases, beautiful UI, citation guardrails, packet export, and tests.

Document Map

1. **Problem and timing**
2. **Product definition**
3. **MVP workflow**
4. **Design direction**
5. **Technical architecture**
6. **AI and evidence safety**
7. **Compliance posture**
8. **Codex tokenmax plan**
9. **GTM and validation**
10. **Source log**

1. Problem and timing

Precise problem: Medicare Advantage post-acute denials are frequently under-appealed because SNF, IRF, LTCH, and hospital discharge teams cannot rapidly assemble the right clinical evidence, map it to the payer's stated reason, and produce a clean reconsideration packet before operational pressure wins.

This is not a vague healthcare AI idea. It is a document-and-workflow problem with a measurable output: a source-cited appeal packet.

Signal	Fact	Why it matters	Source
MA scale	52.8M prior authorization determinations in 2024; 4.1M denied in full or part.	Large recurring workflow surface.	KFF
Under-appeal	Only 11.5% of denied prior authorization requests were appealed.	Pain exists between denial and appeal execution.	KFF
Overturn signal	More than 80% of appealed denials were overturned in 2019-2024.	Appeals can be worth operational investment.	KFF
SNF wedge	OIG found 12% SNF admission denial rate in June 2024 sample; 18% appealed; 95% overturned when appealed.	Precise post-acute wedge with sharp signal.	HHS OIG
Regulatory pressure	CMS-0057-F moves prior authorization toward policy/API reform by 2026/2027.	Design API-ready later, PDF-ready now.	CMS

2. Product definition

AppealForge PAC is a B2B denial recovery workflow product for post-acute providers. It starts after a denial, when the plan has already stated the reason and the provider has clinical documentation.

Input	System action	Output
Denial letter	Parse plan, service, denial reason, deadline, appeal route.	Confirmed denial finding with citations.
Clinical PDFs	Classify pages and extract source quotes.	Evidence map with document/page/quote.
Missing documentation	Create checklist tasks by criterion.	Staff action list.
Reviewed evidence	Draft packet only from cited items.	Appeal letter, evidence table, attachment index.
Decision result	Track outcome and value.	ROI and payer pattern history.

3. MVP workflow

- **Create case:** User enters facility, plan, service type, request type, denial date, and synthetic patient label.
- **Upload documents:** Denial letter and clinical PDFs are uploaded, classified, and extracted into page-level records.

- **Review denial parser:** User confirms denial reason, deadline, payer rationale, and expedited language.
- **Build evidence map:** The system maps denial reasons to criteria and finds supporting source quotes.
- **Resolve gaps:** Missing evidence becomes a targeted checklist instead of fabricated text.
- **Generate packet:** The packet is drafted from reviewed evidence and marked draft.
- **Approve/export:** A human reviewer approves and exports a versioned packet.
- **Track outcome:** Staff record sent date, channel, confirmation, decision, and value.

4. Design direction

The design must not feel like a bland admin portal. The product handles stressful, deadline-driven work, so the interface should feel calm, high-trust, and precise.

Design pillar	Execution	Avoid
Premium clinical command center	Dark shell, warm paper packet, quiet teal/amber signals.	Generic white CRUD.
Evidence-first UX	Source drawer, citation chips, confidence rails.	AI magic claims.
Deadline calm	Subtle amber urgency and next actions.	Alarmist red overload.
Human review trust	Draft states, approval gates, audit stream.	Auto-submit language.

5. Technical architecture

- **Frontend:** Next.js 15, TypeScript, Tailwind, custom AppealForge design tokens.
- **Database:** PostgreSQL + Prisma with organization/facility scoped records.
- **Documents:** Object storage abstraction; local adapter for demo; page-level extraction records.
- **Workers:** Async extraction, parser, evidence, packet, and export jobs.
- **AI:** Structured outputs only; deterministic mock in demo; citation-first production pipeline.
- **Export:** Server-side packet PDF/HTML export with version history.
- **Security:** RBAC, audit logs, no PHI in logs, encryption posture.

6. AI and evidence safety

The central technical moat is not an appeal letter generator. It is the evidence discipline: every claim must be supported or blocked.

Rule	Implementation
No uncited clinical claims	Evidence item requires document id, page number, and quote.
Missing evidence is a task	Create evidence gap with suggested owner and priority.
Draft until approved	AI output starts in `AI_DRAFT`; export requires reviewer gates.
Unsupported claim blocker	Every packet sentence is classified as supported, metadata, template, or blocked.

No proprietary criteria misuse

Use public/general or customer-provided licensed criteria only.

7. Compliance posture

The demo must use synthetic or de-identified data. Production use with PHI requires BAAs, risk analysis, access controls, audit logging, encryption, retention policies, and PHI-safe vendor settings.

- BAA with hosting/storage/OCR/LLM/logging vendors before live PHI.
- No PHI in application logs, analytics, prompts cache, or error monitoring.
- Organization/facility scoped RBAC.
- Audit every PHI-related action.
- Human approval before packet export/submission.
- Data retention and deletion controls.

8. Codex tokenmax plan

The Codex kit is built to prevent a shallow 14-minute scaffold. It forces a living progress file, repeated checks, polish passes, tests, and end-to-end demo flow.

Prompt	Use	Goal
MASTER_TOKENMAX_PROMPT.md	System source of truth.	Keep Codex aligned with product/design/safety.
SPRINT_00_COMBINED_ULTRA.md	One aggressive run.	Build foundation through polish and tests.
SPRINT_01-06	Controlled phases.	Recover if combined sprint stops early.
CONTINUE_PROMPT.md	Resume.	Continue from progress log without repeating.
REVIEW_POLISH_PROMPT.md	Quality pass.	Stress test UI, buyer value, safety, and tests.

9. GTM and validation

Do not overbuild before validation. The first proof is a de-identified historical packet test with regional post-acute operators.

- 20 discovery conversations in 14 days.
- 5 qualified operators with recurring MA denials.
- 3 de-identified packet tests.
- One paid pilot target at \$1,500-\$3,000 per facility per month.
- Pass criteria: recurring denials, painful appeal assembly, appeal wins, budget owner, willingness to pilot.

10. Source log

Key sources reviewed for this kit:

Synthetic/demo planning document · human review required before production PHI

- **KFF 2026:** <https://www.kff.org/medicare/medicare-advantage-insurers-made-nearly-53-million-prior-authorization-determinations-in-2024/>
- **HHS OIG SNF MA denials 2026:** <https://oig.hhs.gov/reports/all/2026/medicare-advantage-organizations-overtaken-nearly-all-appealed-prior-authorization-denials-for-skilled-nursing-facility-admission-raising-concerns-about-initial-denials/>
- **CMS CMS-0057-F:** <https://www.cms.gov/initiatives/burden-reduction/overview/interoperability/policies-regulations/cms-interoperability-prior-authorization-final-rule-cms-0057-f>
- **eCFR 42 CFR 422.568:** <https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-B/part-422/subpart-M/section-422.568>
- **CMS Part C Reconsideration:** <https://www.cms.gov/medicare/appeals-grievances/managed-care/reconsideration-advantage-health-plan-part-c>
- **HHS HIPAA Cloud Computing:** <https://www.hhs.gov/hipaa/for-professionals/special-topics/health-information-technology/cloud-computing/index.html>

Final caution: AppealForge PAC should be sold and built as an evidence assembly workflow, not an autonomous clinical or legal decision system.